SKYNETIX SYSTEMS

n8n
COMPLETE TRAINING BOOK

From workflow automation fundamentals to AI agents, RAG, integrations and production operations

---

Concepts • step-by-step workflows • expressions • JavaScript • AI • 100 hands-on labs

A classroom-ready guide for beginners, developers and automation builders

Version 1.0 | November 2026

Prepared for SkyNetix Systems training classes

How to use this book

This book teaches n8n as a practical automation platform. Learners build workflows progressively: triggers, data transformation, branching, APIs, webhooks, databases, error handling, AI integrations, retrieval workflows, agentic patterns and deployment. The labs use fictional data and safe test endpoints.

Important: n8n changes frequently. Node names, credential screens, node parameters, AI node capabilities and licensing can differ by release and hosting option. Treat the workflow logic as the learning objective and verify exact UI labels in the documentation for your installed version.

Suggested learning path

Phase	Topics	Suggested duration
1. Foundations	Install, UI, workflows, triggers, execution data	4–6 hours
2. Core automation	Mapping, expressions, conditions, loops, code	8–10 hours
3. Integrations	HTTP, webhooks, APIs, databases, files	8–10 hours
4. Reliability	Errors, retries, credentials, testing, scaling	6–8 hours
5. AI workflows	LLMs, structured output, tools, RAG, agents	10–14 hours
6. Capstone	Build, document, secure and present a solution	6–8 hours

Learning outcomes

• Build event-driven workflows with triggers, actions and data mapping.

• Use expressions and JavaScript to transform JSON items.

• Connect REST APIs, webhooks, databases and SaaS services.

• Design reliable workflows with error handling, retries and idempotency.

• Use AI model nodes, structured outputs, tools and retrieval patterns.

• Apply credentials, least privilege, testing, logging and deployment practices.

Table of contents

1. Introduction to workflow automation and n8n

2. Installation, hosting options and first workflow

3. Understanding the editor, nodes, connections and executions

4. Triggers: manual, schedule, webhook and app events

5. Data model: items, JSON, binary data and mapping

6. Expressions and dynamic values

7. Core logic: IF, Switch, Merge, Filter and Loop

8. JavaScript Code node fundamentals

9. HTTP Request node and REST APIs

10. Webhooks: receiving and responding to events

11. Working with files, CSV, spreadsheets and email

12. Databases and persistence

13. SaaS integrations, OAuth and credentials

14. Error handling, retries and workflow reliability

15. Workflow design patterns and reusable sub-workflows
16. Security, permissions and safe automation
17. AI and LLM fundamentals inside n8n
18. Prompt design and structured AI output
19. AI Agent concepts, tools and human approval
20. RAG: document ingestion and retrieval
21. Multi-step AI workflows and agentic automation
22. Testing, debugging, monitoring and performance
23. Deployment, backups, scaling and operations
24. Capstone projects, assessments and glossary

1. Introduction to workflow automation and n8n

Chapter objective: Understand automation, workflows, nodes, triggers and the role of n8n.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. List five repetitive business processes and rank them by frequency, effort and risk.

Lab 2. Draw a workflow for a website enquiry: receive form → validate → notify sales → record lead.

Lab 3. Create a glossary of trigger, node, connection, execution, credential and workflow.

Lab 4. Compare a manual process, a fixed automation and an AI-assisted workflow. Identify where AI is not required.

Expected deliverable

A workflow diagram and an automation opportunity assessment.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

2. Installation, hosting options and first workflow

Chapter objective: Choose a safe learning environment and build a first workflow.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Explore the current n8n Cloud trial/plan information and self-hosting documentation. Record the differences relevant to your class.

Lab 2. If self-hosting locally, follow the official Docker or npm installation guide for your operating system. Do not expose the editor publicly.

Lab 3. Create a workflow with Manual Trigger and Edit Fields (or the equivalent current node) to create name, course and status fields.

Lab 4. Execute the workflow manually, inspect the output and save the workflow with a descriptive name.

Expected deliverable

A first workflow and a short note describing the selected hosting approach.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

3. Understanding the editor, nodes, connections and executions

Chapter objective: Read workflow data and understand how executions are produced.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Create a three-node workflow: Manual Trigger → Edit Fields → No Operation/placeholder. Run it and inspect each node's input and output.

Lab 2. Rename nodes with action-oriented names such as 'Build Lead Record' rather than leaving generic labels.

Lab 3. Run the workflow with two different sample inputs and compare execution data.

Lab 4. Create a workflow description and add notes explaining the purpose of each node.

Expected deliverable

A clearly named and documented workflow with inspected execution data.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

4. Triggers: manual, schedule, webhook and app events

Chapter objective: Select triggers according to the event that should start an automation.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Use Manual Trigger to test a workflow that creates a mock daily report.

Lab 2. Configure a Schedule Trigger to run at a suitable test interval. Verify timezone and avoid unnecessary high-frequency schedules.

Lab 3. Create a Webhook-triggered workflow and send a test JSON request using curl or Postman.

Lab 4. Research one app trigger available in your installed n8n version and document the event, required credential and resulting data.

Expected deliverable

Four trigger examples, with one tested webhook.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

5. Data model: items, JSON, binary data and mapping

Chapter objective: Understand n8n items and how data moves between nodes.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Create three sample customer items using Edit Fields and inspect the JSON output.

Lab 2. Use a mapping node to rename fullName to customerName and normalize an email field.

Lab 3. Create a workflow that processes multiple items and verify whether downstream nodes run once per item or operate on all items.

Lab 4. Download or create a small sample file and inspect how binary data differs from JSON fields.

Expected deliverable

A data transformation workflow with before-and-after examples.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

6. Expressions and dynamic values

Chapter objective: Use expressions to map data dynamically instead of hard-coding values.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Map a customer's name from a previous node into a message body using the expression editor.

Lab 2. Build a greeting expression that handles a missing first name with a fallback.

Lab 3. Use a date/time expression to create a timestamped report title. Verify timezone behavior.

Lab 4. Reference data from a named earlier node and document what happens if that node did not execute.

Expected deliverable

An expression-driven workflow with null-safe fields and documented mappings.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

7. Core logic: IF, Switch, Merge, Filter and Loop

Chapter objective: Route and combine data according to business rules.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Use IF to classify enquiries as high priority when urgency is 'high'. Test both branches.

Lab 2. Use Switch to route support tickets to billing, technical or general support.

Lab 3. Use Merge to combine customer information with an enrichment result. Test matching and unmatched records.

Lab 4. Process a list of records in batches or loop using the appropriate current node pattern. Demonstrate how the workflow stops.

Expected deliverable

A branching workflow with tests for every route and an explanation of item pairing.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

8. JavaScript Code node fundamentals

Chapter objective: Use code only when built-in nodes and expressions are not sufficient.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Use the Code node to normalize whitespace and lowercase an email address for each incoming item.

Lab 2. Create a Code node that calculates a total from quantity and unitPrice and returns a new JSON item.

Lab 3. Filter invalid records and return a separate error list rather than silently dropping data.

Lab 4. Write a code node that summarizes an array of orders. Explain the input and output shape and test empty arrays.

Expected deliverable

A Code-node workflow with readable JavaScript and tested edge cases.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

9. HTTP Request node and REST APIs

Chapter objective: Connect to external REST APIs and understand methods, headers, query parameters and responses.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Call a public GET endpoint using HTTP Request. Set a timeout if supported and inspect status and response body.

Lab 2. Call a test API with query parameters and map two response fields into a new item.

Lab 3. Send a POST request to a safe test endpoint with a JSON body. Do not send real customer data.

Lab 4. Handle a non-2xx response and document status code, error output and retry behavior.

Expected deliverable

A documented API integration with GET and POST examples.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

10. Webhooks: receiving and responding to events

Chapter objective: Receive HTTP events, validate payloads and return predictable responses.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Create a webhook that accepts a JSON enquiry and inspect the incoming headers, query and body.

Lab 2. Validate required fields such as name and email. Return a clear client error for invalid payloads.

Lab 3. Configure a response path using the current Respond to Webhook pattern or response settings.

Lab 4. Test duplicate submissions and design a simple idempotency key strategy.

Expected deliverable

A webhook endpoint with validation, response behavior and duplicate-event handling.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

11. Working with files, CSV, spreadsheets and email

Chapter objective: Automate document and communication tasks with sample data.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Create a CSV of fictional leads and import it into a workflow using the appropriate file/spreadsheet nodes.

Lab 2. Transform CSV rows into clean lead records and reject rows with missing required fields.

Lab 3. Generate a simple summary report and write/export it to a file or spreadsheet.

Lab 4. Configure an email integration in a test environment. Send only to an address you control and include a manual approval step before sending.

Expected deliverable

A lead-import and report workflow with a safe email test.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

12. Databases and persistence

Chapter objective: Store and retrieve workflow state with appropriate database operations.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Connect to a local or training database using a least-privilege account and a synthetic dataset.

Lab 2. Insert a new lead record and verify it using a read query.

Lab 3. Update a lead's status based on a workflow event. Ensure the update targets the correct record.

Lab 4. Implement a duplicate check using a unique lead identifier or idempotency key.

Expected deliverable

A CRUD-style database workflow with duplicate protection.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

13. SaaS integrations, OAuth and credentials

Chapter objective: Authenticate securely to external services and keep secrets out of workflow data.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Review credential types supported by one integration you plan to teach. Record required scopes and permissions.

Lab 2. Create a credential for a sandbox/test account and verify connection without exposing the secret in node output.

Lab 3. Use OAuth where supported and document the authorization and token refresh flow at a high level.

Lab 4. Create a credential inventory showing owner, purpose, scope, environment and rotation/revocation procedure.

Expected deliverable

A working sandbox credential and a credential register.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

14. Error handling, retries and workflow reliability

Chapter objective: Design for transient failures, permanent errors and partial success.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Create a workflow that intentionally fails on one test item and inspect the execution failure details.

Lab 2. Use an error workflow or the current error-handling feature to notify a test channel when a workflow fails.

Lab 3. Add a bounded retry strategy for a simulated temporary API failure. Avoid retrying permanent validation errors.

Lab 4. Design an idempotent processing flow so a repeated event does not create duplicate records or send duplicate notifications.

Expected deliverable

A failure-tested workflow with retry and duplicate-handling behavior.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

15. Workflow design patterns and reusable sub-workflows

Chapter objective: Build workflows that are modular, maintainable and easy to reuse.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Split a lead-processing workflow into intake, validation and notification sub-workflows where appropriate.

Lab 2. Define the input and output contract for a reusable sub-workflow.

Lab 3. Build a parent workflow that calls the sub-workflow and handles its success or failure response.

Lab 4. Create a reusable notification workflow with configurable recipient, subject and message fields.

Expected deliverable

A modular parent/child workflow design with documented contracts.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

16. Security, permissions and safe automation

Chapter objective: Protect credentials, data, endpoints and consequential actions.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Create a threat model for an automation that reads a form and sends an email. Identify trust boundaries and sensitive data.

Lab 2. Use a test credential with the minimum permissions required. Demonstrate that a broader credential is not needed.

Lab 3. Validate webhook payloads and restrict allowed fields and values before downstream use.

Lab 4. Insert a manual approval gate before a simulated payment, deletion or external email. Demonstrate rejection and approval.

Expected deliverable

A security checklist and an automation with least privilege and approval.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

17. AI and LLM fundamentals inside n8n

Chapter objective: Understand how model nodes fit into workflows and how to configure them safely.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Review AI-related nodes available in your n8n version and identify the provider credential requirements.

Lab 2. Create a workflow that sends a short synthetic text to an LLM and captures the response.

Lab 3. Add a prompt that summarizes a fictional support ticket without inventing missing facts.

Lab 4. Compare a normal deterministic workflow with an LLM-assisted workflow and identify which steps should remain deterministic.

Expected deliverable

A simple AI workflow and a short model/provider configuration note.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

18. Prompt design and structured AI output

Chapter objective: Create prompts that produce predictable output suitable for downstream automation.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Write a prompt that classifies a fictional support ticket into billing, technical or general.

Lab 2. Require JSON output with category, priority, summary and confidence. Parse and validate the output.

Lab 3. Add a fallback route for invalid JSON, missing fields or unsupported category values.

Lab 4. Create a test set of ten tickets and measure classification accuracy against expected labels.

Expected deliverable

A validated classification workflow with a small evaluation set.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

19. AI Agent concepts, tools and human approval

Chapter objective: Build a bounded AI agent workflow with controlled tools.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Create an agent workflow that can use two safe tools: read-only FAQ lookup and calculator.

Lab 2. Write clear tool descriptions and restrict tool inputs. Test an unknown tool request.

Lab 3. Add a maximum-step or bounded execution strategy where available and document how the workflow exits.

Lab 4. Add human approval before any write action such as sending an email or updating a customer record.

Expected deliverable

An agent with limited tools, tested boundaries and human approval.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

20. RAG: document ingestion and retrieval

Chapter objective: Ground AI responses in approved documents rather than model memory alone.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Create three fictional policy documents and load them into an ingestion workflow.

Lab 2. Split documents into chunks and preserve metadata such as document ID, title and department.

Lab 3. Generate embeddings using a supported embedding integration and store them in a vector store supported by your setup.

Lab 4. Retrieve relevant chunks for a question and ask the model to answer only from the retrieved context, citing document IDs.

Expected deliverable

A small RAG pipeline with traceable source metadata and no-answer behavior.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

21. Multi-step AI workflows and agentic automation

Chapter objective: Combine deterministic routing, model decisions, tools and human review.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Build a support workflow: intake → validate → classify → retrieve policy → draft response.

Lab 2. Add a route for low-confidence or missing-evidence cases to human review.

Lab 3. Create a multi-step research workflow with separate collect, summarize and verify stages. Keep the verifier independent.

Lab 4. Add a final audit record containing workflow run ID, selected route, tool calls and approval status without logging secrets.

Expected deliverable

A complete agentic business process with evidence, review and audit trail.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

22. Testing, debugging, monitoring and performance

Chapter objective: Validate workflows with repeatable tests and monitor operational health.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Create test fixtures for valid, invalid, empty and duplicate input payloads.

Lab 2. Use pinned sample data to reproduce a workflow execution and document how to inspect node input/output.

Lab 3. Create a monitoring workflow that records success/failure counts and execution duration from available operational data.

Lab 4. Measure a baseline execution time and then reduce unnecessary nodes or repeated API calls. Record the trade-off.

Expected deliverable

A test matrix, troubleshooting notes and basic performance report.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

23. Deployment, backups, scaling and operations

Chapter objective: Prepare a workflow environment for controlled, supportable use.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Document a development, test and production environment separation strategy.

Lab 2. Review official backup and restore guidance for your hosting model. Perform a restore drill in a non-production environment.

Lab 3. Document how credentials, encryption keys, database backups and environment variables are managed for your deployment.

Lab 4. Create an operations runbook covering upgrades, monitoring, failed executions, rollback, access reviews and incident response.

Expected deliverable

A deployment diagram, backup/restore evidence and operations runbook.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

24. Capstone projects, assessments and glossary

Chapter objective: Deliver a complete automation that solves a real business problem safely.

Classroom setup: Use a separate workflow for each lab or clearly named sections in a single practice workflow. Save sample input and screenshots/exports of successful and failed executions.

Hands-on labs

Lab 1. Capstone A — Lead management: webhook intake, validation, deduplication, CRM/database storage, routing and sales notification.

Lab 2. Capstone B — AI support assistant: ticket intake, classification, RAG over policy documents, draft answer and human approval.

Lab 3. Capstone C — Recruitment automation: candidate form intake, structured data extraction, duplicate detection, status tracking and recruiter review. Use synthetic candidate data.

Lab 4. Capstone D — Content operations: schedule a content brief, generate a draft, run quality checks and require approval before publication.

Expected deliverable

A working capstone, exported workflow, README, architecture diagram, test evidence and demo.

Instructor checkpoint

Ask the learner to explain the trigger, data shape, node sequence, credentials used, failure behavior and how the result was verified. Require one positive test and one negative or edge-case test.

Stretch challenge

Add a new business rule, test it with at least three input examples, document the change and confirm that previously working paths still behave correctly.

Appendix A. JavaScript and expression quick reference

Common expression patterns

```
{{ $json.email }}
{{ $json.firstName || 'there' }}
{{ $json.amount * $json.quantity }}
{{ $now }}
{{ $('Webhook').first().json.body.email }}
```

Expressions are evaluated by n8n's expression engine. The exact available helpers and data access methods can vary by context and version. Use the expression editor's suggestions and the current documentation.

Code node: illustrative item transformation

```
// Example pattern for a Code node.
// Check the current Code node mode and return contract for your n8n version.
return $input.all().map(item => {
  const email = String(item.json.email ?? '').trim().toLowerCase();

  return {
    json: {
      ...item.json,
      email,
      isValidEmailShape: /^[^\\s@]+@[^\\s@]+\\.[^\\s@]+$/.test(email),
    },
  };
});
```

Debugging checklist

• Inspect the input and output of the node that first produces unexpected data.

• Check whether the value is in JSON or binary data.

• Check whether the node receives one item or multiple items.

• Check null, missing and differently typed fields.

• Check credential scopes, endpoint, HTTP method and response status.

• Re-run with a small, known sample payload.

Appendix B. Common workflow design patterns

Event intake: Trigger → validate → normalize → deduplicate → persist → acknowledge.

Scheduled reporting: Schedule → fetch → aggregate → format → review → deliver.

API orchestration: Trigger → call service A → transform → call service B → reconcile.

Human approval: Prepare draft → request approval → IF approved → perform action; otherwise cancel.

AI classification: Receive text → classify → validate schema → route; unknown/invalid → human review.

RAG assistant: Ingest → chunk → embed → store; query → retrieve → ground prompt → answer with sources.

Reliable processing: Idempotency key → bounded retry → error workflow → audit record.

Appendix C. Assessment rubric and student submission

Criteria	Weight	Evidence
Workflow correctness	20%	Expected path works for valid inputs
Data handling	15%	Correct mapping, types, JSON and binary data
Integrations	15%	API/authentication used appropriately
Reliability	15%	Errors, retries and duplicates handled
AI quality	15%	Structured output, grounding and fallback
Security	10%	Least privilege, secret handling, approval
Documentation/demo	10%	Clear workflow, README and explanation

Student submission checklist

Submit 1. Workflow export or reproducible workflow instructions are included.

Submit 2. README explains purpose, trigger, setup, credentials and execution.

Submit 3. No secrets or real confidential data are included.

Submit 4. At least five test cases cover success and failure paths.

Submit 5. Human approval is shown for consequential actions.

Submit 6. Known limitations and operational owner are documented.

Appendix D. Official documentation and learning references

n8n documentation: https://docs.n8n.io/

n8n courses: https://docs.n8n.io/courses/

n8n integrations: https://n8n.io/integrations/

n8n community: https://community.n8n.io/

n8n GitHub: https://github.com/n8n-io/n8n

Docker documentation: https://docs.docker.com/

Postman learning center: https://learning.postman.com/

JavaScript guide (MDN): https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide

OpenAI API documentation: https://platform.openai.com/docs/

Model Context Protocol: https://modelcontextprotocol.io/

Licensing and commercial use note: Review n8n's current license and commercial-use terms before offering hosted n8n services or embedding it in a paid product. Do not assume that self-hosting automatically grants every form of commercial redistribution or white-label use.